

★ WEEK 11 · PART 2 · PRACTICE

The OU Process, Worked by Hand

Fit Mean Reversion, Then Trade It

From an AR(1) coefficient to a half-life and a spread trade

Part 2's mean-reversion model, with the arithmetic shown. We fit an Ornstein-Uhlenbeck process to a spread through its AR(1) coefficient - reading off a half-life and equilibrium band - then turn those into entry, exit, and an expected holding time.

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PART 2 · APPLICATIONS

Estimate Discretely, Interpret Continuously

Case A: fit OU to a spread via its AR(1) coefficient - mapping to the reversion speed, half-life, and equilibrium band.
Case B: trade that spread - entry and exit from the band, holding time from the half-life. Then a Claude Code lab.

What Part 2 Told Us

Now we fit and trade it - with numbers

1

OU is Brownian motion with a leash

The drift pulls the process back toward a mean θ at speed κ . Unlike GBM, it doesn't wander off - it reverts.

2

The half-life measures the pull

How long to close half the gap to the mean: $\ln 2$ over κ . It's the natural timescale of the reversion.

3

OU is a continuous AR(1)

Fit an AR(1) to the discrete data, map its coefficient to κ , and interpret the result in continuous time.

Everything here is a calculator exercise: one AR(1) coefficient, one half-life, one band.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a fitted AR(1) coefficient.
- Map to kappa, half-life, and the band.
- Then turn those into a trade.

What you should watch

- How the coefficient becomes a reversion speed.
- How the half-life sets the timescale.
- How the band sets entry and exit.

Fit OU to a Spread

From a coefficient to a continuous model

A pairs spread reverts to zero. We fit a simple AR(1) to the daily data and get a coefficient of 0.90. Because OU is a continuous AR(1), that one number maps to the reversion speed kappa, from which we read the half-life and the equilibrium band - a full continuous-time characterization estimated discretely

The inputs

- Daily AR(1) coefficient $\phi = 0.90$.
- Innovation $sd = 0.30$, mean $\theta = 0$.
- Sampling interval one day.

What we'll do by hand

- Map ϕ to the reversion speed κ .
- Compute the half-life.
- Compute the equilibrium band.

OU and Its AR(1) Bridge

One process, two representations

$$\begin{aligned} \text{OU: } dX &= \kappa(\theta - X) dt + \sigma dW && \text{(a pull toward } \theta) \\ \text{discretized: } X_t &= \phi X_{t-1} + \dots, \quad \phi = e^{-\kappa \Delta t} && \text{(OU is a continuous AR(1))} \end{aligned}$$

In continuous time

- The drift pulls toward theta at speed kappa.
- Sigma is the size of the shocks.
- That's the OU stochastic differential equation.

On a grid

- Sampled daily, OU becomes an AR(1).
- Its coefficient is e-to-the-minus-kappa.
- So the coefficient carries all the reversion.

Speed and Half-Life

From the coefficient to the timescale

$$\kappa = -\frac{\ln \phi}{\Delta t} = -\ln 0.90 = 0.105/\text{day}, \quad \text{half-life} = \frac{\ln 2}{\kappa} = \frac{0.693}{0.105} = \mathbf{6.6} \text{ days}$$

The mapping

- kappa is minus the log of the coefficient.
- A coefficient of 0.90 gives kappa 0.105.
- Higher persistence, slower reversion.

The half-life

- $\ln 2$ divided by kappa is 6.6 days.
- That's the time to close half the gap.
- The natural clock of the spread.

The Equilibrium Band

Where the spread lives

$$\text{stationary sd} = \frac{\sigma_\varepsilon}{\sqrt{1-\phi^2}} = \frac{0.30}{\sqrt{1-0.81}} = \mathbf{0.69}, \quad \text{band } \pm 2\text{sd} = \pm 1.38$$

The stationary sd

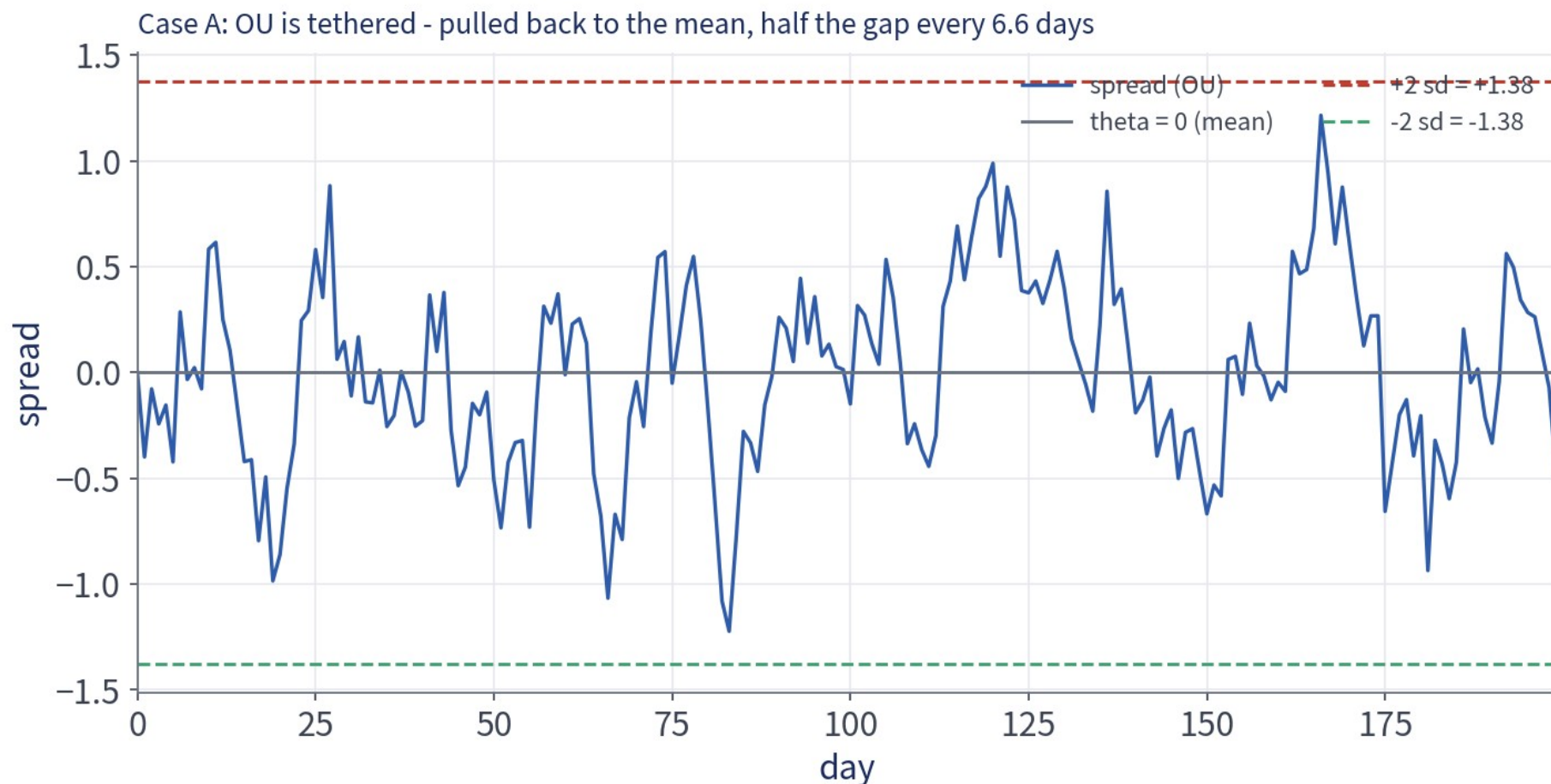
- Innovation sd over root of one-minus-phi-squared.
- That's 0.30 over root 0.19, about 0.69.
- It's how far the spread typically strays.

The band

- Two standard deviations is plus or minus 1.38.
- The spread sits inside this band most of the time.
- It becomes the trigger for a trade.

Tethered, Not Wandering

Unlike Brownian motion, the OU spread is pulled back whenever it strays - closing half the gap to the mean every 6.6 days. It spends most of its time inside the plus-or-minus-1.38 band, which is exactly what makes it tradable



Estimate Discretely, Interpret Continuously

The bridge that makes OU practical

Why the bridge matters

- You never observe continuous time - only samples.
- AR(1) is trivial to fit to the daily data.
- The mapping recovers the continuous parameters.

What you get

- A reversion speed with real-world units.
- A half-life you can act on.
- An equilibrium band for trading.

Trade the OU Spread

Entry, exit, and holding time. **With OU fitted, the trade writes itself. Enter when the spread reaches the edge of its band - two standard deviations from the mean - betting it reverts. The expected gap then decays exponentially, halving every half-life, so the holding time follows directly. We compute the reversion path and the expected hold by**

The plan

- Enter at plus or minus 1.38.
- Exit as the spread nears the mean.
- The half-life sets the timescale.

What we'll do by hand

- Set the entry and exit thresholds.
- Track the expected gap decaying.
- Read off the expected holding time.

Entry and Exit From the Band

The band becomes the trigger

enter at $|X| \geq 2 \text{sd} = 1.38$ (short if high, long if low), exit near $X = \theta = 0$

The entry

- When the spread hits two sd, it's stretched.
- Short it if high, buy it if low.
- Bet on the pull back to the mean.

The exit

- Close as the spread returns near theta.
- The reversion is the profit.
- The band sizes both ends of the trade.

The Reversion, Halving

How the gap closes over time

$$\mathbb{E}[X_t] = X_0 e^{-\kappa t} \Rightarrow \text{the gap halves every half-life : } 2 \text{ sd} \rightarrow 1 \text{ sd} \rightarrow 0.5 \text{ sd}$$

The decay

- The expected gap follows e-to-the-minus-kappa-t.
- Each half-life halves what remains.
- From 2 sd to 1 sd to 0.5 sd.

Reading it

- One half-life: down to one sd.
- Two: down to half a sd.
- The pull is fast at first, then eases.

Expected Holding and Profit

How long, and how much

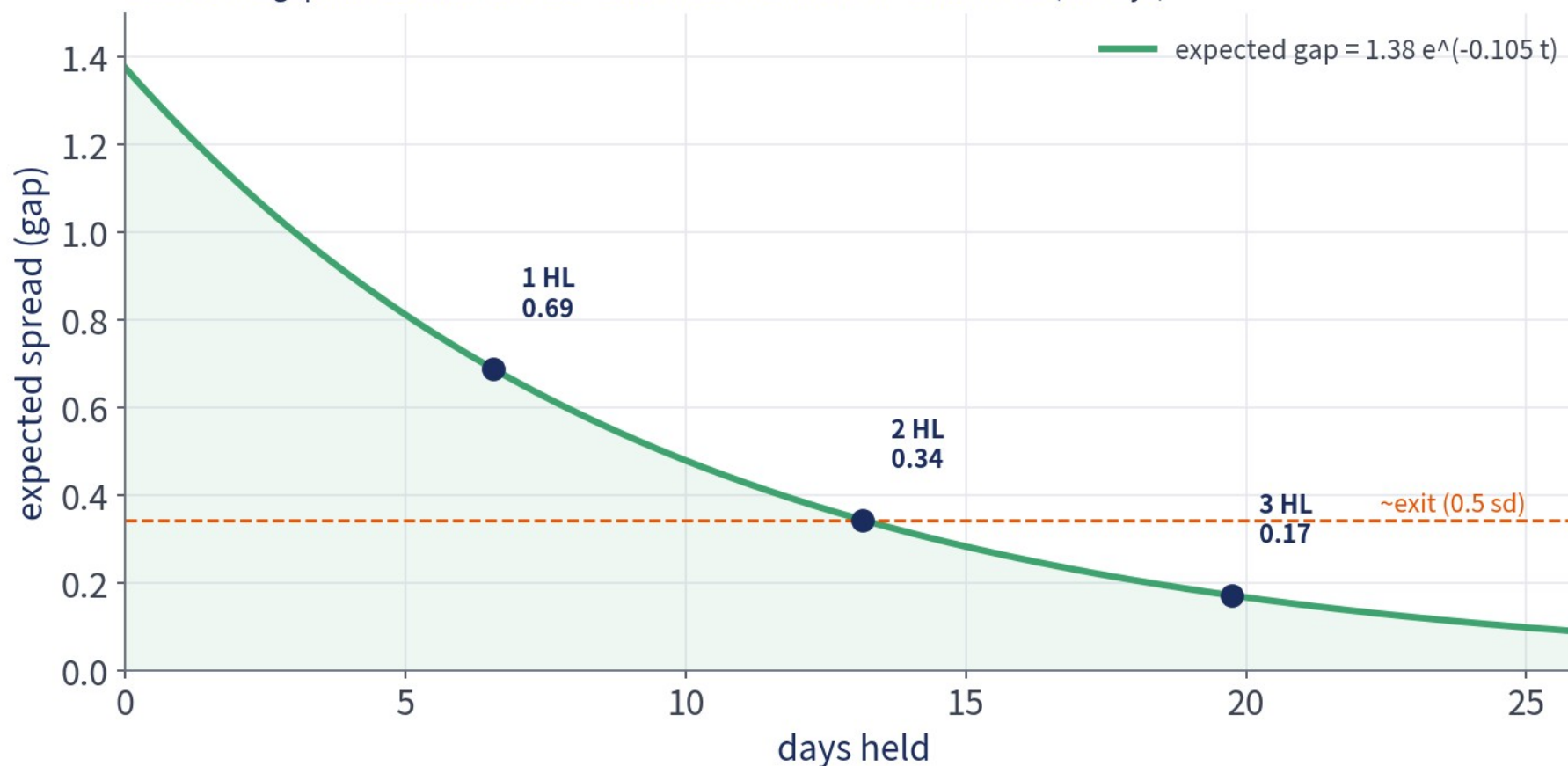
from 2 sd to $\sim 0.25\text{ sd}$: ≈ 3 half-lives = **20** days (expected holding)
profit per trade $\approx$ captured reversion = 1.38 spread units (before costs)

From two sd back toward the mean takes about three half-lives - some 20 days. The captured reversion, about 1.38 spread units, is the gross profit before costs.

The Gap Halves Each Half-Life

Enter at the top of the band (1.38) and the expected spread decays exponentially: one half-life brings it to 1 sd, two to 0.5 sd, three to about 0.25 sd - near the exit. The half-life is the clock that tells you how long your capital is committed

Case B: the gap halves each half-life - back near the mean in ~3 half-lives (20 days)



The Half-Life Runs the Trade

Timescale and thresholds from one fit

What the fit gives you

- The band tells you when to enter.
- The half-life tells you how long to wait.
- Both from a single AR(1) coefficient.

Why it's disciplined

- A slow reverter ties up capital longer.
- A wide band means bigger, rarer trades.
- The parameters set the whole trade profile.

Tethered Randomness, Made Tradable

Fit, then trade

Both cases turn on one idea: OU is mean reversion you can measure. Fit an AR(1), map it to a reversion speed, and you have a half-life and an equilibrium band. Those two numbers are the entire trade - when to enter, how far the move is worth, and how long you'll wait for it. Estimate discretely, interpret and trade

The fit (Case A)

- AR(1) coefficient to reversion speed.
- Half-life of 6.6 days.
- Equilibrium band of plus or minus 1.38.

The trade (Case B)

- Enter at the band, exit at the mean.
- Gap halves every half-life.
- Held about 20 days for 1.38 units.

Reproduce the Fit and the Trade

Direct Claude Code to check your arithmetic

GOAL Verify the OU fit and the reversion decay, then backtest the spread trade.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - fit OU:

- * from AR(1) $\phi=0.90$, innovation $sd=0.30$, $dt=1$,
print $\kappa=-\ln(\phi)$, the half-life, the
stationary sd , and the ± 2 sd band;

CASE B - trade it:

- * set entry at 2 sd , exit at the mean;
- * print the expected gap at 1, 2, 3 half-lives;
- * report the expected holding time.

Then fit OU to a real spread and backtest the trade.

Match every printed number to the slides.

Check against the slides

- $\kappa = 0.105$ per day.
- half-life = 6.6 days.
- stationary $sd = 0.69$, band ± 1.38 .
- hold ~ 20 days.

Deliverable: `part2_practice.py` whose printed numbers match the OU fit and the reversion decay.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you get kappa from an AR(1) coefficient?

**kappa = minus the
log of the coefficient**

Q2 What is the half-life?

**$\ln 2$ over kappa - time
to close half the gap**

Q3 How does the expected gap decay over time?

**Exponentially -
halving every half-
life**

All three come from the AR(1)-to-OU mapping and the reversion decay.

Three Pitfalls to Avoid

Where OU trading slips

Common mistakes

- "Read kappa straight off the coefficient."
- "Assume the spread reverts on a fixed schedule."
- "Trust the fit when the relationship has changed."

The reality

- kappa is minus the log of the coefficient, not one minus it.
- Reversion is exponential in expectation, not a countdown.
- Re-estimate - a broken spread has no half-life.

Two Calculations to Remember

Part 2 applications in one screen

1

Fit OU through its AR(1) coefficient

A coefficient of 0.90 maps to a reversion speed κ of 0.105 per day, a half-life of 6.6 days, and a stationary sd of 0.69 - an equilibrium band of plus or minus 1.38. A full continuous model, estimated from daily data.

2

Trade it from the band and the half-life

Enter at two sd (1.38) and the expected gap halves every half-life: 2 sd to 1 sd to 0.5 sd. Reverting near the mean takes about three half-lives - some 20 days - to capture roughly 1.38 spread units.

3

One fit sets the whole trade

The band says when to enter, the reversion says how far, and the half-life says how long. Estimate discretely, interpret and trade continuously - the OU bridge in one workflow.

Next — Part 3 practice: OU in trading and Stage V, where mean reversion completes the pairs strategy.